

Volume 13

Trust Architecture & Security Intelligence

Version: 1.0

Status: Draft

Chapter 1

Vision

Traditional security platforms generate alerts.

Project Sentinel shall generate **trust**.

Everything inside the platform contributes to one objective:

Can this object be trusted today?

Not yesterday.

Not last month.

Today.

Chapter 2

Universal Trust Model

Every object receives four independent evaluations.

Identity Trust

Together these form the object's **Trust Profile**.

Chapter 3

Identity Trust

Questions answered:

- Is this the expected user?
 - Is MFA enabled?
 - Has identity changed?
 - Is the account dormant?
 - Is privilege appropriate?
 - Are credentials healthy?
-

Chapter 4

Configuration Trust

Questions answered:

- Is configuration compliant?
 - Is encryption enabled?
 - Are unnecessary services disabled?
 - Are patches current?
 - Is hardening applied?
-

Chapter 5

Behavior Trust

Questions answered:

- Is activity normal?
- Has behavior changed significantly?
- Are new startup mechanisms present?
- Are unusual network patterns observed?
- Has software changed unexpectedly?

Behavior should always be assessed against evidence collected from authorized sources.

Chapter 6

Exposure Trust

Questions answered:

- Internet facing?
 - Open ports?
 - Weak TLS?
 - Public buckets?
 - Exposed APIs?
 - Weak authentication?
-

Chapter 7

Unified Trust Profile

Instead of dozens of disconnected scores:

Trust Profile

Each percentage must be backed by explainable evidence.

Chapter 8

Explainable Security

This is a fundamental engineering requirement.

Every score must answer:

Why?

How?

Evidence?

Confidence?

Recommendations?

Limitations?

If the platform cannot explain a score, it should not present it.

Chapter 9

Threat Context

The platform shall distinguish between:

- Configuration weaknesses.
- Active indicators.
- Environmental risk.
- Historical issues.
- Confirmed incidents.

This helps avoid unnecessary alarm.

Chapter 10

Security Confidence

Every conclusion receives a confidence rating.

Example

Malware

or
Persistence Detected

This reinforces our evidence-first philosophy.

Chapter 11

Trust Timeline

Trust changes over time.
January

Users can see whether security is improving or declining.

Chapter 12

Trust Heatmap

A visual overview of organizational trust.
Example
Identity

This supports quick decision-making.

PS-ADR-0010

Evidence Before Conclusions

Status: Accepted

Decision

Project Sentinel shall not make definitive security claims without sufficient supporting evidence.

All conclusions shall include:

- Evidence
- Confidence
- Explanation
- Recommended actions
- Known limitations

Rationale

Evidence-based reporting improves user trust, reduces false positives, and supports auditability.

New Core Engine

Adaptive Security Engine (ASE)

One idea I believe can become a major differentiator is the **Adaptive Security Engine**.

Instead of relying only on static rules, the engine builds a baseline of the user's authorized environment.

It answers questions such as:

- What is normal for this laptop?
- What is normal for this website?
- What is normal for this API?
- What is normal for this employee's device?

It then highlights meaningful deviations and explains why they matter. This is not "AI magic"; it is evidence-based behavioral comparison with transparent reasoning.

New Core Engine

Security Knowledge Engine 2.0

We previously introduced the Knowledge Engine. I now propose expanding it into a comprehensive internal knowledge system.

It will maintain relationships between:

- Assets
- Vulnerabilities
- Findings
- Threat intelligence
- Security controls
- Compliance requirements
- Incidents
- Evidence
- Remediation actions
- Reports

This allows the platform to answer questions like:

"Which critical web applications depend on an endpoint that recently lost compliance?" without the user manually correlating data.

New Core Engine

Decision Intelligence Engine (DIE)

This engine does **not** replace human judgment.

Its purpose is to prioritize.

For every finding, it estimates:

- Security impact.
- Operational impact.
- Estimated remediation effort.
- Dependencies.
- Business criticality.

The output is a ranked action plan supported by evidence.

A New Long-Term Goal: Digital Security Advisor

This is where I believe Project Sentinel can become unique.

Most platforms stop after showing findings.

Project Sentinel should continue by acting as a **Digital Security Advisor**.

For example:

"Your organization reduced its external attack surface by 18% over the last three months. The next action with the highest security benefit is enabling MFA for the remaining administrative accounts. Based on your current environment, this is expected to significantly improve identity protection."

Notice what is **not** happening:

- No exaggerated claims.
- No unexplained AI.

- No fear-based messaging.

Only evidence, context, and recommendations.

A strategic observation before we continue

Looking at everything we have designed so far, I believe Project Sentinel is evolving into three integrated layers:

1. **Cyber Security Operating Platform** – the technical foundation (engines, objects, workflows, APIs).
2. **Cyber Security Intelligence Platform** – correlation, knowledge, evidence, reasoning, and decision support.
3. **Cyber Security Experience Platform** – an interface that makes advanced security understandable and actionable for everyone.

Those three layers working together—not any single scanning engine—are what can make Project Sentinel genuinely distinctive. Our next volumes will define the remaining engines (web assessment, endpoint assessment, mobile assessment, cloud assessment, compliance, reporting, plugin SDK, APIs, and testing) with the same engineering rigor so that the Blueprint becomes a complete, implementable specification.

====[USER]=====

Good and continue

====[ASSISTANT]=====

Excellent. I think we should now define something that most cybersecurity products are missing:

A cybersecurity operating system should think in terms of missions, knowledge, and decisions—not just scans.

This is where Project Sentinel begins to differentiate itself.

PROJECT SENTINEL BLUEPRINT

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